

- 1 A student takes measurements to determine the acceleration of a ball as it rolls down a slope. He uses the apparatus illustrated in Fig. 1.1.

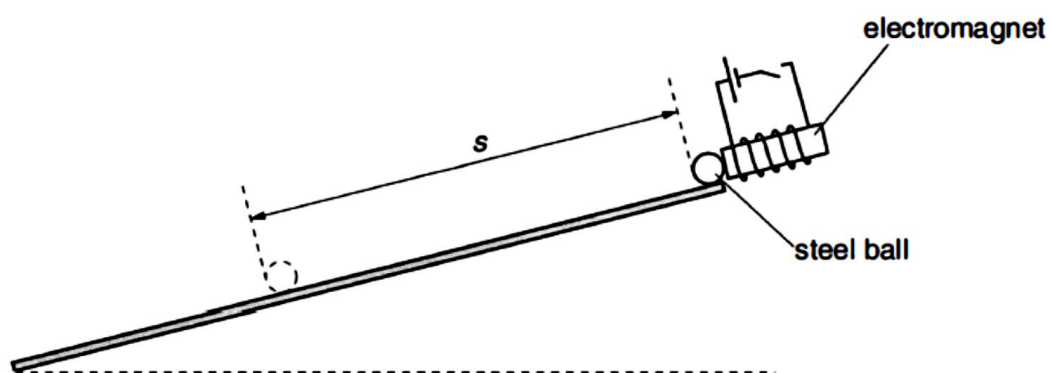


Fig. 1.1

The student measures the time t for the ball to roll a distance s down the slope after the ball has been released from the electromagnet.

The variation with t^2 of the distance is shown in Fig. 1.2.

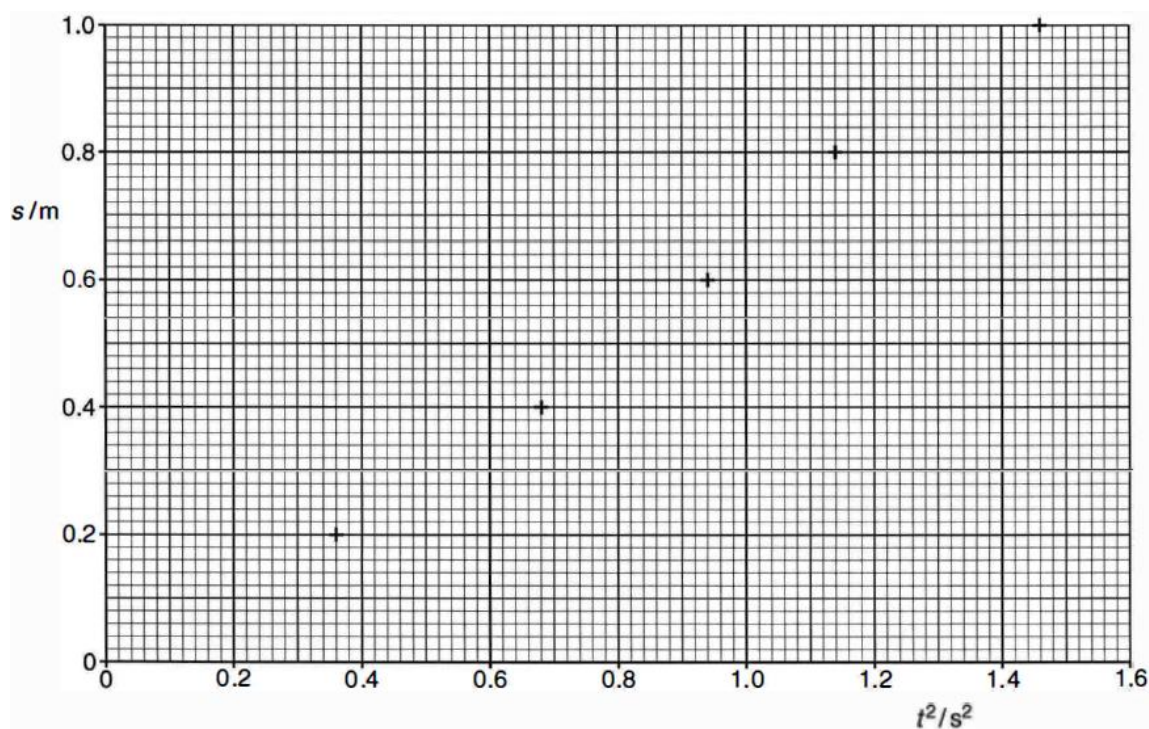
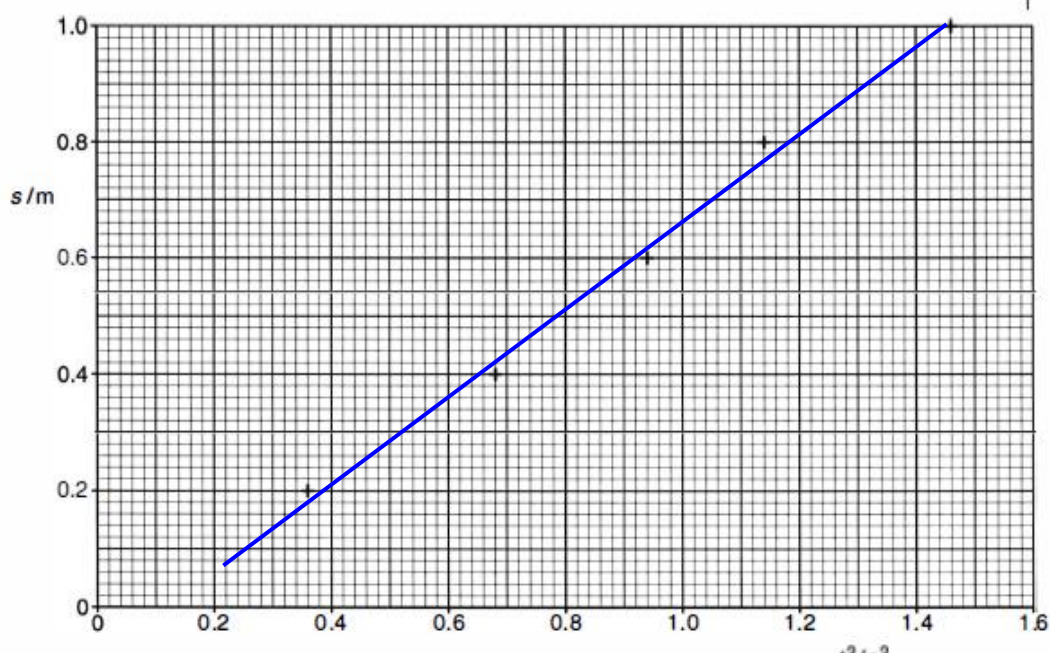


Fig. 1.2

- (i) Use Fig. 1.2 to determine a value for the acceleration of the ball down the slope.



BFL

Using $s = ut + \frac{1}{2}at^2$ and initial velocity is zero, the gradient of graph is $\frac{1}{2}a$.

Using (0.40, 0.22) and (1.20, 0.82) to find gradient,

$$\text{gradient} = \frac{0.82-0.22}{1.20-0.40} = 0.75 \text{ [M1]} (0.73 \rightarrow 0.77)$$

Hence, acceleration = $0.75 \times 2 = 1.50 \text{ ms}^{-2}$.

acceleration =m s⁻² [5]

(ii) State the feature of the data shown in Fig. 1.2 that indicates the presence of

1. random error,

The data points are scattered about the line of best fit.

[1]

2. and systematic error.

The best-fit line does not pass through the origin since the ball has zero displacement when the ball starts to move.

[1]

2	Fig. 2.1 shows a block of mass $M = 4.0 \text{ kg}$ released from a vertical height of 6.0 m on a curved